

Research Process

Presented By:

Dr. Prakash Datt Bhatt

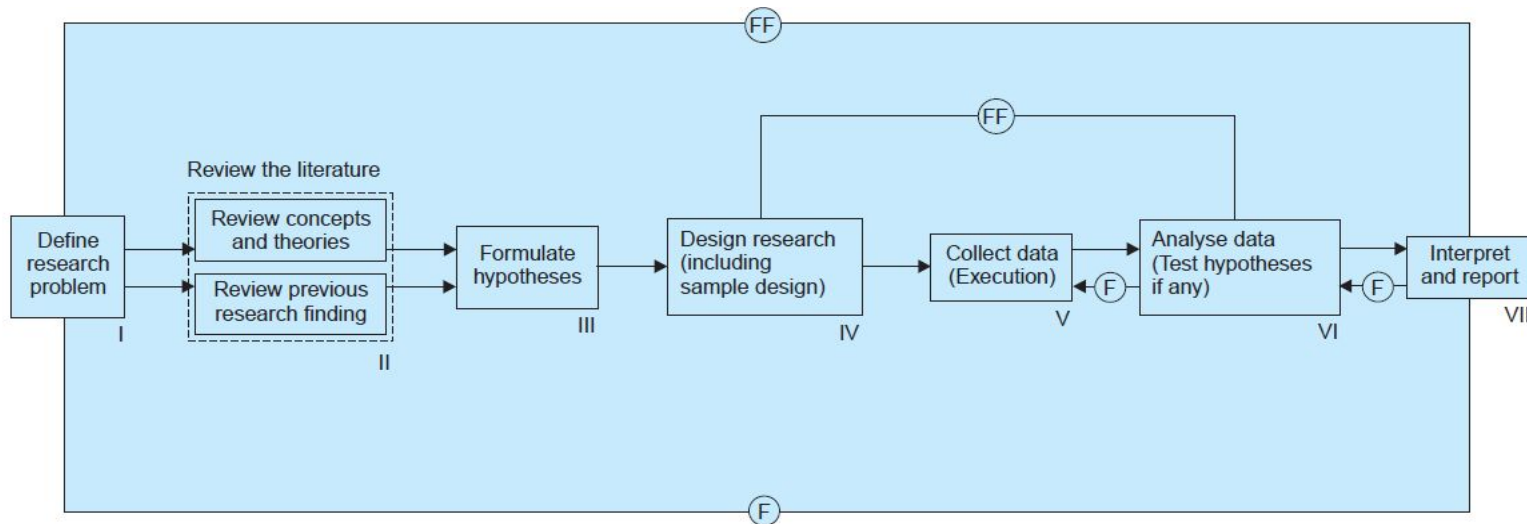
prakash.bhatt@sncs.tu.edu.np

Department of Computer Science

Siddhanath Science Campus, Mahendranagar

What is Research Process?

- Research process consists of series of actions or steps necessary to effectively carry out research and the desired sequencing of these steps.
- The figure illustrates a research process



Where (F) = feed back (Helps in controlling the sub-system to which it is transmitted)
(FF) = feed forward (Serves the vital function of providing criteria for evaluation)

What is Research Process?

- The figure indicates that the research process consists of a number of closely related activities, however, these activities overlap continuously rather than following a strictly prescribed sequence.
- At times, the first step determines the nature of the last step to be undertaken.

Steps in Research Process

- If subsequent procedures have not been taken into account in the early stages, serious difficulties may arise which may even prevent the completion of the study.
- One should remember that the various steps involved in a research process are not mutually exclusive; nor they are separate and distinct.
- They do not necessarily follow each other in any specific order and the researcher has to be constantly anticipating at each step in the research process the requirements of the subsequent steps.

Steps in Research Process

- However, the following order concerning various steps provides a useful procedural guideline regarding the research process:
 - Formulating the research problem
 - Extensive literature survey
 - Developing the hypothesis
 - Preparing the research design
 - Determining sample design;
 - Collecting the data
 - Execution of the project
 - Analysis of data
 - Hypothesis testing;
 - Generalizations and interpretation
 - Preparation of the report or presentation of the results,

Formulating the Research Problem

- There are two types of research problem, viz., those
 - Relate to states of nature
 - Relationship between variables.

- Essentially two steps are involved in define research problem, viz.,
 - understanding the problem thoroughly and
 - rephrasing the same into meaningful terms from an point of view.

Formulating the Research Problem

- The best way of understanding the problem is to discuss it with one's own colleagues or with those having some expertise in the matter.
- In an academic institution the researcher can seek the help from a guide who is usually an experienced man and has several research problems in mind.
- After this the researcher rephrases the problem into analytical or operational terms i.e., to put the problem in as specific terms as possible.

Formulating the Research Problem

- This task of formulating, or defining, a research problem is a step of greatest importance in the entire research process.
- The problem to be investigated must be defined unambiguously for that will help discriminating relevant data from irrelevant ones.

Literature Review

- Once the problem is define, a brief summary of it should be written down. It is compulsory for a research worker writing a thesis for a Ph.D. degree to write a synopsis of topic and submit it to necessary committee or the research board for approval.
- At this juncture the researcher should undertake extensive literature survey connected with the problem. For this purpose, the abstracting and indexing journals and published or unpublished bibliographies are the first place to go to.

Literature Review

- Academic journals, conference proceedings, government reports, books etc., must be tapped depending on the nature of the problem.
- In this process, it should be remembered that one source will lead to another.
- The earlier studies, if any, which are similar to the study in hand should be carefully studied. A good library will be a great help to the researcher at this stage.

Formulate Hypothesis

- Formulate hypothesis is tentative assumption made in order to draw out and test its logical or empirical consequences.
- Hypothesis should be very specific and limited to the piece of research in hand because it has to be tested.
- The role of the hypothesis is to guide the researcher by delimiting the area of research and to keep him on the right track.

Research Design

- The function of research design is to provide for the collection of relevant evidence with minimal expenditure of effort, time and money.

- Research purpose may be grouped into four categories, viz.,
 - Exploration
 - Description
 - Diagnosis
 - Experimentation.

Data Collection

- Primary data can be collected through:
 - By Observation
 - Through personal interview
 - Through telephone interview
 - By mailing of questionnaires
 - Through Schedules

References

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